IBM Z DATATHON 2026 - THEME 2: AI SECURED

# AegisNet

## One fraud model across banks.

## No raw customer data ever leaves a bank.

Local differential privacy + federated learning + secure aggregation, merged on an IBM LinuxONE (s390x) VM.

**0.833**

federated AUC

**0.744-0.822**

each bank alone

**3 of 3**

banks, zero raw rows shared

WHY BANKS LOSE TO FRAUD RINGS

# The problem

### Fraud crosses banks

Criminals spread attacks over several banks. Each bank sees only its own slice, so patterns look harmless in isolation.

### Data cannot be pooled

Customer transactions are protected by privacy law and bank policy (for example GDPR, PCI-DSS). A shared data lake is not an option.

### Each bank is blind

In our data Bank A mostly sees account takeover, Bank B cross-border fraud, Bank C card testing. Alone, each misses the others.

**Question: can banks learn from each other's fraud without sharing a single raw transaction?**

AegisNet says yes, and shows measured proof.

THREE PROTECTIONS, ONE SHARED MODEL

# The solution

**1 Local privacy**

Each bank clips features to public bounds, scales to [0,1] and adds Laplace noise (scale 1/ε). ε is the privacy dial.

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**2 Train locally**

Each bank trains logistic regression on its own noisy data. Raw rows never leave the bank.

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**3 Mask updates**

Banks add pairwise random masks to their updates. Masks cancel in the sum, so the server sees only noise per bank.

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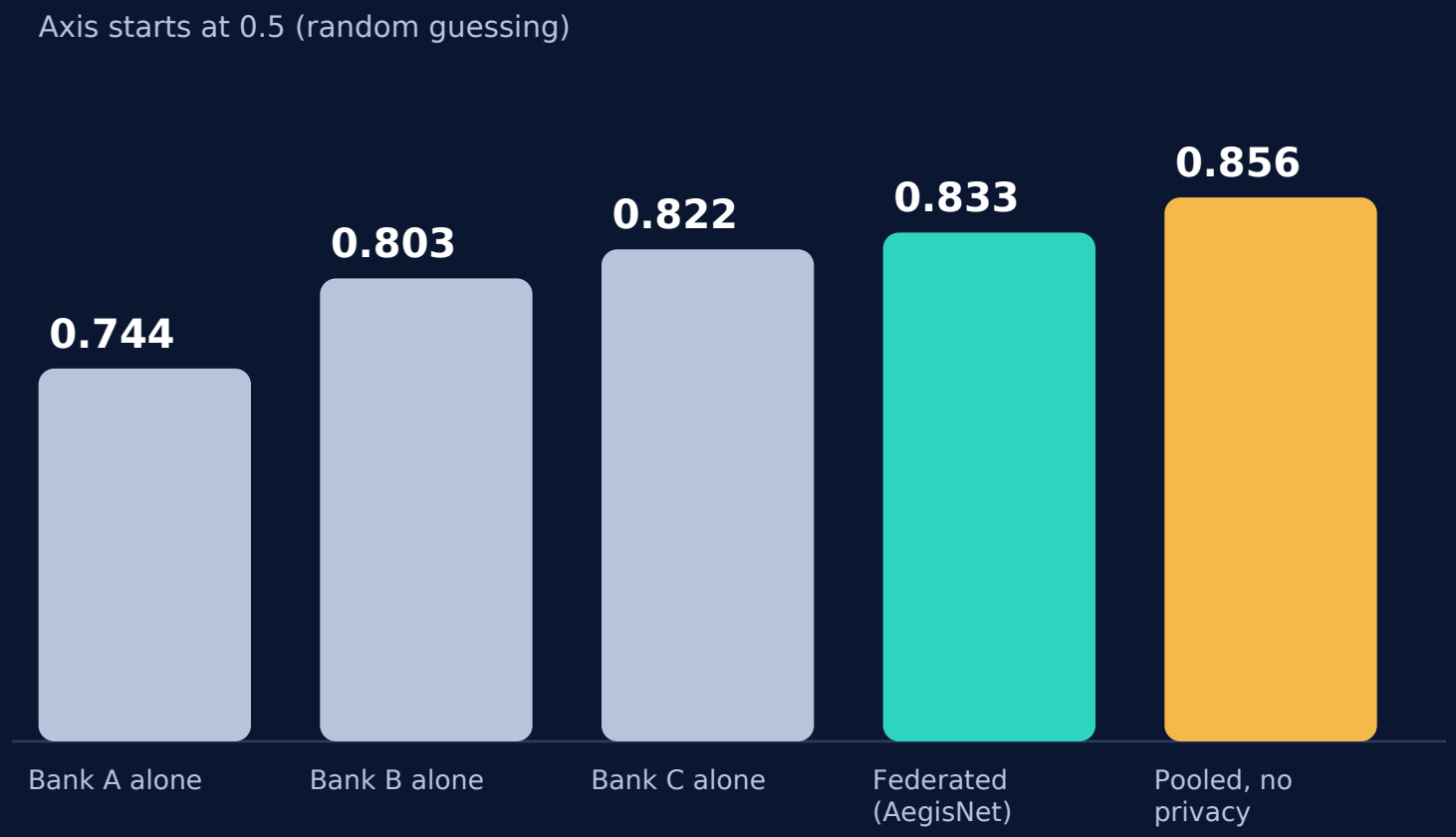
**4 Merge on LinuxONE**

FedAvg over 10 rounds gives the exact merged model. The shared model is sent back to all banks.

Design choice: logistic regression, not Random Forest, because its weights can be averaged. A forest stays as a local baseline.

SYNTHETIC DATA | ε = 5 | 10 ROUNDS | AUC ON UNSEEN HOLDOUT

# Measured results




## What it shows

Federated beats every bank alone (0.833 vs 0.744-0.822) and gets close to the unsafe pooled ceiling (0.856). Sweeping ε from 0.5 to no noise moves federated AUC from about 0.71 to 0.85. Masks cancel exactly.

THE TRUSTED PLACE TO MERGE

# Why IBM Z / LinuxONE

## What we ran on LinuxONE

The pipeline runs on an IBM LinuxONE VM (s390x, Ubuntu 24.04): data generation, privacy, federated training, secure-aggregation check and the Streamlit dashboard. The proof script prints the AUC table and confirms the masks cancel.

## Why it fits

The merge step is the one place every bank must trust. LinuxONE brings high availability, strong isolation and throughput for a shared fraud service, which is what a consortium of banks needs.

## Planned integration (not claimed as done)

Real attestation so banks verify the aggregator before sending updates, using Secure Execution / Hyper Protect where the environment offers it. Faster scoring with on-chip AI acceleration on z16-class machines. Hardware features are claimed only after checking the machine.

DASHBOARD AND SCENARIO PAGES

# Built to be shown

### Overview and Federation

Animated flow diagram and a live ε slider. Banks, noise and AUC update together so judges can see the trade-off.

### Alerts console

Scores international test transactions with the federated model, then a simulated block, freeze and PII-free case file. Simulation only.

### Laundering scanner

Animated route maps of six public cases and a rule-based scanner on a synthetic ledger. Catches 8 of 8 planted schemes, 0 false positives.

Plus a LinuxONE page that reads real machine facts and says plainly when it is not on s390x.

BUSINESS CASE

# Value and who benefits

### Banks

Catch fraud patterns they cannot see alone, without a data-sharing agreement over raw records and without a new privacy risk.

### Customers

Fewer fraudulent transactions and fewer wrongly frozen cards, while their data stays inside their own bank.

### Regulators

A privacy dial (ε), masked updates and an auditable process that supports GDPR and PCI-DSS style obligations.

**Pitch in one line: each bank misses two of three fraud patterns alone; together they see all three, with privacy by design.**

WHAT IS REAL TODAY, WHAT COMES NEXT

# Honest limits and roadmap

## Limits today

Data is synthetic. Secure-aggregation masks are simulated in one process (no real key exchange or dropout recovery). ε is applied per feature, so per-record privacy loss is larger than ε. Class labels are not noised. The AML scanner is rule-based and does not use the federated model yet.

## Roadmap

Real attestation of the aggregator. Membership-inference attack demo with and without differential privacy. A real fraud dataset split across simulated banks. Malicious-bank test with robust aggregation (median or trimmed mean). Auditor page with privacy budget per round and an exportable compliance report.

THANK YOU

# AegisNet

## Each bank keeps its data.
## Together they catch more fraud.

Code, notebooks and run instructions are in the public GitHub repository linked in the submission.

Open-source stack: Streamlit, NumPy, pandas, scikit-learn, SciPy, Plotly.